

SuSiE-relax: Prototype and Lite Variational Inference

Computation note of the susie-relax model.

Throughout, $x \in \mathbb{R}^J$ is the vector of marginal summary statistics, N is the GWAS sample size, and $\bar{R}$ is a reference LD correlation matrix. We write

$$\Omega = \bar{R}^{-1}, \quad \mu_j = \bar{R}_{-j,j}, \quad S_j = \bar{R}_{-j,-j} - \bar{R}_{-j,j}\bar{R}_{j,-j}.$$

For $z_j \in \mathbb{R}^{J-1}$, let $u_j(z_j) \in \mathbb{R}^J$ be the vector with j -th entry equal to 1 and remaining entries equal to z_j .

1 SER-relax-prototype

SER-relax-prototype is the original single-effect regression model with uncertainty in the active LD column. The model is

$$x \mid \beta, \gamma = j, d_j, z_j \sim \mathcal{N}_J \left(\beta d_j u_j(z_j), \frac{\bar{R}}{N} \right), \quad (1)$$

$$\beta \sim \mathcal{N}(0, \sigma_0^2), \quad (2)$$

$$\gamma \sim \text{Unif}\{1, \dots, J\}. \quad (3)$$

The relaxed LD column is induced by an inverse-Wishart model for the unknown correlation matrix R ,

$$R \sim IW_J(\nu_0 \bar{R}, \nu_0 + J + 1).$$

For a fixed column j , define $d_j = R_{jj}$ and $z_j = R_{-j,j}/R_{jj}$. The induced column model is

$$d_j \sim \text{IG} \left(\frac{\nu_0 + 2}{2}, \frac{\nu_0}{2} \right), \quad (4)$$

$$z_j \mid \omega_j \sim \mathcal{N}_{J-1} \left(\mu_j, \frac{S_j}{\omega_j} \right), \quad (5)$$

$$\omega_j \sim \text{Gamma} \left(\frac{\nu_0 + 3}{2}, \frac{1}{2} \right), \quad (6)$$

using the shape–rate parameterization for Gamma distributions. Larger ν_0 concentrates R around $\bar{R}$; smaller ν_0 allows the active LD column to move farther away from the reference LD.

1.1 Variational Inference

We use the mean-field approximation

$$q(\gamma = j, \beta, d_j, z_j, \omega_j) = \pi_j q_j(\beta) q_j(d_j) q_j(z_j) q_j(\omega_j),$$

where

$$q_j(\beta) = \mathcal{N}(m_{\beta j}, v_{\beta j}), \quad q_j(z_j) = \mathcal{N}_{J-1}(m_{zj}, V_{zj}),$$

and

$$q_j(\omega_j) = \text{Gamma}(\alpha_{\omega j}, \beta_{\omega j}).$$

Let

$$\bar{\beta}_j = m_{\beta j}, \quad \overline{\beta^2}_j = v_{\beta j} + m_{\beta j}^2, \quad \bar{d}_j = \mathbb{E}_j[d_j], \quad \overline{d^2}_j = \mathbb{E}_j[d_j^2], \quad \bar{\omega}_j = \mathbb{E}_j[\omega_j].$$

Define

$$A_j = 1 + \text{tr}(S_j^{-1} V_{zj}) + (m_{zj} - \mu_j)^\top S_j^{-1} (m_{zj} - \mu_j), \quad (7)$$

$$B_j = x_j + (m_{zj} - \mu_j)^\top S_j^{-1} (x_{-j} - \mu_j x_j), \quad (8)$$

$$Q_j = \text{tr}(S_j^{-1} V_{zj}) + (m_{zj} - \mu_j)^\top S_j^{-1} (m_{zj} - \mu_j). \quad (9)$$

Then the Gaussian update for the effect size is

$$v_{\beta j} = \left(\sigma_0^{-2} + N \overline{d^2}_j A_j \right)^{-1}, \quad (10)$$

$$m_{\beta j} = v_{\beta j} N \bar{d}_j B_j. \quad (11)$$

Let

$$T_{1j} = \bar{\beta}_j \bar{d}_j, \quad T_{2j} = \overline{\beta^2}_j \overline{d^2}_j.$$

The Gaussian update for the relaxed LD column is

$$V_{zj} = \frac{S_j}{N T_{2j} + \bar{\omega}_j}, \quad (12)$$

$$m_{zj} = \mu_j + \frac{N T_{1j}}{N T_{2j} + \bar{\omega}_j} (x_{-j} - \mu_j x_j). \quad (13)$$

The Gamma factor for ω_j is

$$\alpha_{\omega j} = \frac{\nu_0 + J + 2}{2}, \quad (14)$$

$$\beta_{\omega j} = \frac{1 + Q_j}{2}. \quad (15)$$

The factor $q_j(d_j)$ is non-standard. Its optimal density is

$$q_j(d_j) \propto d_j^{-(\nu_0+2)/2-1} \exp \left\{ -\frac{\nu_0}{2d_j} - \frac{N}{2} \overline{\beta^2}_j A_j d_j^2 + N \bar{\beta}_j B_j d_j \right\}, \quad d_j > 0.$$

In implementation we work with $r_j = \log d_j$. The unnormalized log-density in r_j , including the Jacobian, is

$$\ell_j(r) = - \left(\frac{\nu_0 + 2}{2} + 1 \right) r - \frac{\nu_0}{2e^r} - \frac{N}{2} \overline{\beta^2}_j A_j e^{2r} + N \bar{\beta}_j B_j e^r + r.$$

Numerical quadrature over r gives $\mathbb{E}_j[d_j]$, $\mathbb{E}_j[d_j^2]$, $\mathbb{E}_j[d_j^{-1}]$, $\mathbb{E}_j[\log d_j]$, and $E_j[\log q_j(d_j)]$.

The coordinate probability is

$$\pi_j = \frac{\exp(\mathcal{L}_j)}{\sum_{k=1}^J \exp(\mathcal{L}_k)},$$

where the local lower bound is

$$\begin{aligned} \mathcal{L}_j = & \log(1/J) + E_j \log p(x \mid \beta, d_j, z_j, \gamma = j) + E_j \log p(\beta) + E_j \log p(d_j \mid \nu_0) \\ & + E_j \log p(\omega_j \mid \nu_0) + E_j \log p(z_j \mid \omega_j) - E_j \log q_j(\beta) - E_j \log q_j(d_j) \\ & - E_j \log q_j(\omega_j) - E_j \log q_j(z_j). \end{aligned} \quad (16)$$

The full SER-relax-prototype ELBO is

$$\mathcal{L}_{\text{prototype}} = \sum_{j=1}^J \pi_j \{\mathcal{L}_j - \log \pi_j\}.$$

1.2 Learning ν_0

Holding q fixed, the ν_0 -dependent part of the prototype ELBO is

$$M_{\text{prototype}}(\nu_0) = \sum_{j=1}^J \pi_j \{E_j \log p(d_j \mid \nu_0) + E_j \log p(\omega_j \mid \nu_0)\}.$$

With

$$a_d = \frac{\nu_0 + 2}{2}, \quad b_d = \frac{\nu_0}{2}, \quad a_\omega = \frac{\nu_0 + 3}{2}, \quad b_\omega = \frac{1}{2},$$

this objective becomes

$$\begin{aligned} M_{\text{prototype}}(\nu_0) = & \sum_{j=1}^J \pi_j \left[a_d \log b_d - \log \Gamma(a_d) - (a_d + 1) \mathbb{E}_j \log d_j - b_d \mathbb{E}_j(d_j^{-1}) \right. \\ & \left. + a_\omega \log b_\omega - \log \Gamma(a_\omega) + (a_\omega - 1) \mathbb{E}_j \log \omega_j - b_\omega \mathbb{E}_j \omega_j \right]. \end{aligned} \quad (17)$$

This one-dimensional objective can be optimized directly over $\nu_0 > 0$, or over $\eta = \log \nu_0$ with Newton updates. Direct bounded optimization is usually more stable in practice.

1.3 Implementation Algebra

The key identities come from the block inverse of $\bar{R}$. Because $\bar{R}$ is a correlation matrix,

$$\Omega_{-j,-j} = S_j^{-1}, \quad \Omega_{-j,j} = -S_j^{-1} \mu_j.$$

Therefore

$$u_j(z_j)^\top \Omega u_j(z_j) = 1 + (z_j - \mu_j)^\top S_j^{-1} (z_j - \mu_j), \quad (18)$$

$$u_j(z_j)^\top \Omega x = x_j + (z_j - \mu_j)^\top S_j^{-1} (x_{-j} - \mu_j x_j). \quad (19)$$

These identities avoid repeated manipulation of full $J \times J$ columns. The prototype is nevertheless computationally heavier than the main method because it requires quadrature for every $q_j(d_j)$ and a Gamma update for every $q_j(\omega_j)$.

2 SER-relax

SER-relax is the simplified single-effect model. It keeps the central idea that the active LD column can relax around $\bar{R}$, but removes the two latent variables that made the prototype slow:

$$d_j = 1, \quad \omega_j = \nu_0 + 3.$$

Equivalently, with

$$\tau^2 = (\nu_0 + 3)^{-1}, \quad \tau^{-2} = \nu_0 + 3,$$

the relaxed-column prior is

$$z_j \mid \nu_0 \sim \mathcal{N}_{J-1}(\mu_j, \tau^2 S_j).$$

The single-effect model is

$$x \mid \beta, \gamma = j, z_j \sim \mathcal{N}_J\left(\beta u_j(z_j), \frac{\bar{R}}{N}\right), \quad (20)$$

$$\beta \sim \mathcal{N}(0, \sigma_0^2), \quad (21)$$

$$\gamma \sim \text{Unif}\{1, \dots, J\}. \quad (22)$$

2.1 Variational Inference

We use

$$q(\gamma = j, \beta, z_j) = \pi_j q_j(\beta) q_j(z_j),$$

where

$$q_j(\beta) = \mathcal{N}(m_{\beta j}, v_{\beta j}), \quad q_j(z_j) = \mathcal{N}_{J-1}(m_{zj}, V_{zj}).$$

For a current $q_j(z_j)$, define

$$\bar{\beta}_j = m_{\beta j}, \quad \overline{\beta^2}_j = v_{\beta j} + m_{\beta j}^2,$$

and the same A_j , B_j , and Q_j as above, but without d_j and ω_j . The update for $q_j(\beta)$ is

$$v_{\beta j} = (\sigma_0^{-2} + NA_j)^{-1}, \quad (23)$$

$$m_{\beta j} = v_{\beta j} NB_j. \quad (24)$$

The update for $q_j(z_j)$ is

$$V_{zj} = \frac{S_j}{N\overline{\beta^2}_j + \tau^{-2}}, \quad (25)$$

$$m_{zj} = \mu_j + \frac{N\bar{\beta}_j}{N\overline{\beta^2}_j + \tau^{-2}}(x_{-j} - \mu_j x_j). \quad (26)$$

The local lower bound is

$$\begin{aligned} \mathcal{L}_j^{\text{SER}} &= \log(1/J) + E_j \log p(x \mid \beta, z_j, \gamma = j) + E_j \log p(\beta) + E_j \log p(z_j \mid \nu_0) \\ &\quad - E_j \log q_j(\beta) - E_j \log q_j(z_j). \end{aligned} \quad (27)$$

Thus

$$\pi_j = \frac{\exp(\mathcal{L}_j^{\text{SER}})}{\sum_{k=1}^J \exp(\mathcal{L}_k^{\text{SER}})}.$$

The expected log-likelihood is

$$\begin{aligned} E_j \log p(x \mid \beta, z_j, \gamma = j) &= -\frac{J}{2} \log(2\pi) - \frac{1}{2} \log \left| \frac{\bar{R}}{N} \right| \\ &\quad - \frac{N}{2} \left[x^\top \Omega x - 2\bar{\beta}_j B_j + \overline{\beta^2}_j A_j \right]. \end{aligned} \quad (28)$$

The relaxed-column prior contribution is

$$E_j \log p(z_j \mid \nu_0) = -\frac{J-1}{2} \log(2\pi) - \frac{1}{2} \log |S_j| + \frac{J-1}{2} \log \tau^{-2} - \frac{\tau^{-2}}{2} Q_j.$$

The full SER-relax ELBO is

$$\mathcal{L}_{\text{SER}} = \sum_{j=1}^J \pi_j \{ \mathcal{L}_j^{\text{SER}} - \log \pi_j \}.$$

2.2 Learning ν_0

Holding q fixed, only the Gaussian prior terms for z_j depend on ν_0 . Hence, up to constants,

$$M_{\text{SER}}(\nu_0) = \sum_{j=1}^J \pi_j \left\{ \frac{J-1}{2} \log \tau^{-2} - \frac{\tau^{-2}}{2} Q_j \right\}.$$

Since $\sum_j \pi_j = 1$, the closed-form update in the τ^2 parameterization is

$$\hat{\tau}^2 = \frac{\sum_{j=1}^J \pi_j Q_j}{J-1}, \quad \hat{\nu}_0 = \frac{1}{\hat{\tau}^2} - 3.$$

In code we bound ν_0 to avoid invalid or numerically extreme updates. Since $\nu_0 > 0$, the natural prior variance satisfies $\tau^2 < 1/3$.

2.3 Implementation Algebra

SER-relax can be implemented without explicitly forming all S_j^{-1} matrices. For a vector r , define

$$C_j(r) = (r_{-j} - \mu_j r_j)^\top S_j^{-1} (r_{-j} - \mu_j r_j).$$

The block inverse identity gives

$$C_j(r) = r^\top \Omega r - r_j^2.$$

Also, because $\bar{R}_{jj} = 1$, the Schur determinant identity gives

$$|S_j| = |\bar{R}|.$$

Write

$$z_{\text{denom},j} = N \overline{\beta^2}_j + \tau^{-2}, \quad z_{\text{scale},j} = \frac{N \bar{\beta}_j}{z_{\text{denom},j}}.$$

Then

$$m_{zj} = \mu_j + z_{\text{scale},j}(r_{-j} - \mu_j r_j), \quad V_{zj} = S_j / z_{\text{denom},j}.$$

The quantities needed for the beta update and the local ELBO are

$$Q_j = \frac{J-1}{z_{\text{denom},j}} + z_{\text{scale},j}^2 C_j(r), \quad (29)$$

$$A_j = 1 + Q_j, \quad (30)$$

$$B_j = r_j + z_{\text{scale},j} C_j(r). \quad (31)$$

Thus each coordinate only needs r_j , $C_j(r)$, the shared $\log |\bar{R}|$, and the current scalar variational parameters.

3 SuSiE-relax

SuSiE-relax is the multi-effect version of SER-relax. This is the main method. For effects $\ell = 1, \dots, L$,

$$\gamma_\ell \sim \text{Unif}\{1, \dots, J\}, \quad \beta_\ell \sim \mathcal{N}(0, \sigma_\ell^2),$$

and, conditional on $\gamma_\ell = j$,

$$z_{\ell j} \mid \nu_0 \sim \mathcal{N}_{J-1}(\mu_j, \tau^2 S_j), \quad \tau^{-2} = \nu_0 + 3.$$

The observation model is

$$x \mid \{\beta_\ell, \gamma_\ell, z_{\ell, \gamma_\ell}\}_{\ell=1}^L \sim \mathcal{N}_J \left(\sum_{\ell=1}^L \beta_\ell u_{\gamma_\ell}(z_{\ell, \gamma_\ell}), \frac{\bar{R}}{N} \right).$$

3.1 Variational Inference

We use the independent effect approximation

$$q = \prod_{\ell=1}^L q_\ell(\gamma_\ell) \prod_{j=1}^J q_{\ell j}(\beta_\ell) q_{\ell j}(z_{\ell j}),$$

where

$$q_\ell(\gamma_\ell = j) = \pi_{\ell j}, \quad q_{\ell j}(\beta_\ell) = \mathcal{N}(m_{\beta, \ell j}, v_{\beta, \ell j}),$$

and

$$q_{\ell j}(z_{\ell j}) = \mathcal{N}_{J-1}(m_{z, \ell j}, V_{z, \ell j}).$$

Define the posterior mean contribution of effect ℓ by

$$\bar{\theta}_\ell = \sum_{j=1}^J \pi_{\ell j} \bar{\beta}_{\ell j} u_j(m_{z, \ell j}).$$

When updating effect ℓ , use the current residual

$$r_\ell = x - \sum_{k \neq \ell} \bar{\theta}_k.$$

For each coordinate j , define

$$A_{\ell j} = 1 + \text{tr}(S_j^{-1} V_{z,\ell j}) + (m_{z,\ell j} - \mu_j)^\top S_j^{-1} (m_{z,\ell j} - \mu_j), \quad (32)$$

$$B_{\ell j}(r_\ell) = r_{\ell j} + (m_{z,\ell j} - \mu_j)^\top S_j^{-1} (r_{\ell,-j} - \mu_j r_{\ell j}), \quad (33)$$

$$Q_{\ell j} = \text{tr}(S_j^{-1} V_{z,\ell j}) + (m_{z,\ell j} - \mu_j)^\top S_j^{-1} (m_{z,\ell j} - \mu_j). \quad (34)$$

Let

$$\bar{\beta}_{\ell j} = m_{\beta,\ell j}, \quad \overline{\beta^2}_{\ell j} = v_{\beta,\ell j} + m_{\beta,\ell j}^2.$$

The updates are the single-effect SER-relax updates applied to the residual r_ℓ :

$$v_{\beta,\ell j} = (\sigma_\ell^{-2} + N A_{\ell j})^{-1}, \quad (35)$$

$$m_{\beta,\ell j} = v_{\beta,\ell j} N B_{\ell j}(r_\ell), \quad (36)$$

$$V_{z,\ell j} = \frac{S_j}{N \bar{\beta^2}_{\ell j} + \tau^{-2}}, \quad (37)$$

$$m_{z,\ell j} = \mu_j + \frac{N \bar{\beta}_{\ell j}}{N \bar{\beta^2}_{\ell j} + \tau^{-2}} (r_{\ell,-j} - \mu_j r_{\ell j}). \quad (38)$$

The coordinate weights are

$$\pi_{\ell j} = \frac{\exp(\mathcal{L}_{\ell j}^{\text{coord}})}{\sum_{k=1}^J \exp(\mathcal{L}_{\ell k}^{\text{coord}})},$$

where $\mathcal{L}_{\ell j}^{\text{coord}}$ is the SER-relax local lower bound computed with residual r_ℓ , prior variance σ_ℓ^2 , and the variational factors for effect ℓ . After updating $\pi_{\ell j}$, recompute $\bar{\theta}_\ell$.

3.2 Full ELBO

Let

$$C_x = -\frac{J}{2} \log(2\pi) - \frac{1}{2} \log \left| \frac{\bar{R}}{N} \right|.$$

The expected log-likelihood is

$$\begin{aligned} E_q \log p(x \mid \cdot) = C_x - \frac{N}{2} \left[x^\top \Omega x - 2 \sum_{\ell=1}^L \bar{\theta}_\ell^\top \Omega x \right. \\ \left. + \sum_{\ell=1}^L \sum_{j=1}^J \pi_{\ell j} \overline{\beta^2}_{\ell j} A_{\ell j} + 2 \sum_{1 \leq \ell < k \leq L} \bar{\theta}_\ell^\top \Omega \bar{\theta}_k \right]. \end{aligned} \quad (39)$$

The full SuSiE-relax ELBO is

$$\begin{aligned} \mathcal{L}_{\text{SuSiE-relax}} = E_q \log p(x \mid \cdot) \\ + \sum_{\ell=1}^L \sum_{j=1}^J \pi_{\ell j} \left[\log(1/J) + E_{\ell j} \log p(\beta_\ell) + E_{\ell j} \log p(z_{\ell j} \mid \nu_0) \right. \\ \left. - E_{\ell j} \log q_{\ell j}(\beta_\ell) - E_{\ell j} \log q_{\ell j}(z_{\ell j}) \right] - \sum_{\ell=1}^L \sum_{j=1}^J \pi_{\ell j} \log \pi_{\ell j}. \end{aligned} \quad (40)$$

Here

$$E_{\ell j} \log p(z_{\ell j} | \nu_0) = -\frac{J-1}{2} \log(2\pi) - \frac{1}{2} \log |S_j| + \frac{J-1}{2} \log \tau^{-2} - \frac{\tau^{-2}}{2} Q_{\ell j}.$$

3.3 Learning ν_0

The ν_0 -dependent part of the SuSiE-relax ELBO is

$$M_{\text{SuSiE-relax}}(\nu_0) = \sum_{\ell=1}^L \sum_{j=1}^J \pi_{\ell j} \left\{ \frac{J-1}{2} \log \tau^{-2} - \frac{\tau^{-2}}{2} Q_{\ell j} \right\}.$$

Because $\sum_j \pi_{\ell j} = 1$ for each effect,

$$\hat{\tau}^2 = \frac{\sum_{\ell=1}^L \sum_{j=1}^J \pi_{\ell j} Q_{\ell j}}{L(J-1)}, \quad \hat{\nu}_0 = \frac{1}{\hat{\tau}^2} - 3.$$

In implementation this update is bounded, for example by $\nu_0 \in [\nu_{\min}, \nu_{\max}]$. Empirically, monitoring both the ELBO and ν_0 is important: with small reference panels, ν_0 can move slowly and apparent duplicated effects may be transient until the relaxation level has converged.

3.4 Implementation Algebra

The same scalar representation used in SER-relax applies to each effect-specific residual r_ℓ . Define

$$C_{\ell j} = C_j(r_\ell) = r_\ell^\top \Omega r_\ell - r_{\ell j}^2.$$

With

$$z_{\text{denom}, \ell j} = N \bar{\beta}_{\ell j}^2 + \tau^{-2}, \quad z_{\text{scale}, \ell j} = \frac{N \bar{\beta}_{\ell j}}{z_{\text{denom}, \ell j}},$$

we have

$$Q_{\ell j} = \frac{J-1}{z_{\text{denom}, \ell j}} + z_{\text{scale}, \ell j}^2 C_{\ell j}, \quad (41)$$

$$A_{\ell j} = 1 + Q_{\ell j}, \quad (42)$$

$$B_{\ell j}(r_\ell) = r_{\ell j} + z_{\text{scale}, \ell j} C_{\ell j}. \quad (43)$$

This is the main computational simplification: each effect update only requires the residual r_ℓ , the quadratic $r_\ell^\top \Omega r_\ell$, and coordinate-wise scalar operations.

The posterior mean effect can also be computed without explicitly forming every $m_{z, \ell j}$. Since

$$u_j(m_{z, \ell j}) = \bar{R}_{\cdot j} + z_{\text{scale}, \ell j} \{r_\ell - r_{\ell j} \bar{R}_{\cdot j}\},$$

the effect mean is

$$\begin{aligned} \bar{\theta}_\ell &= \sum_{j=1}^J \pi_{\ell j} \bar{\beta}_{\ell j} [\bar{R}_{\cdot j} + z_{\text{scale}, \ell j} \{r_\ell - r_{\ell j} \bar{R}_{\cdot j}\}] \\ &= \bar{R} \{ \pi_\ell \odot \bar{\beta}_\ell \odot (1 - z_{\text{scale}, \ell} \odot r_\ell) \} + r_\ell \sum_{j=1}^J \pi_{\ell j} \bar{\beta}_{\ell j} z_{\text{scale}, \ell j}. \end{aligned} \quad (44)$$

This is the formula used by the fast implementation. It avoids building a $J \times J$ matrix of relaxed columns for every effect.

3.5 IBSS Algorithm

The coordinate-ascent procedure is an IBSS-style algorithm:

1. Initialize $\pi_{\ell j}$, $m_{\beta, \ell j}$, $v_{\beta, \ell j}$, $m_{z, \ell j}$, $V_{z, \ell j}$, $\bar{\theta}_\ell$, and ν_0 .
2. For each effect $\ell = 1, \dots, L$, form

$$r_\ell = x - \sum_{k \neq \ell} \bar{\theta}_k.$$

3. For all coordinates j , update $q_{\ell j}(\beta_\ell)$, $q_{\ell j}(z_{\ell j})$, $A_{\ell j}$, $B_{\ell j}$, and $Q_{\ell j}$ using the scalar algebra above.
4. Compute the coordinate lower bounds $\mathcal{L}_{\ell j}^{\text{coord}}$, update $\pi_{\ell j}$ by softmax, and recompute $\bar{\theta}_\ell$.
5. After sweeping through all effects, update the shared ν_0 using the closed-form moment update, with bounds.
6. Compute the full ELBO and repeat until both the ELBO and ν_0 have stabilized.